



NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
2025-2026

Batch Number	CB10
Team Members	Nakka Vijay Bhaskar Reddy (22471A05H8) Nagaran Prasad Rao (22471A05H7) Mundlamuri Vijaya Kumarachari (22471A05H6)
Guide	Dr. Sireesha Moturi, M.Tech., Ph.D., Associate Professor
Title	enhanced multi-zonal forest type classification using yolov8 and eurosat for scalable environmental monitoring
Domain/Technology	DEEP LEARNING
Base Paper Link	https://ceur-ws.org/Vol-3988/paper13.pdf
Dataset Link	https://www.kaggle.com/datasets/apollo2506/eurosat-dataset
Software Requirements	Browser: Any latest browser like Chrome Operating System: Windows 7 Server or later Python (COLAB)
Hardware Requirements	SystemType: Intel Core i5 or above RAM: 8 GB Number of cores:5 Number of Threads: 4
Abstract	Forests are a critical component of the Earth's ecosystem, yet monitoring them accurately over large areas remains a significant challenge. This research presents an intelligent deep learning (DL) approach using satellite imagery to achieve high-accuracy forest stand detection. The study aims to develop a faster, scalable, and automated forest analysis framework compared to traditional field-based surveys, utilizing the EuroSAT dataset for effective land cover classification. The YOLOv8-nano model was trained and evaluated to assess its capability in identifying forest and non-forest regions. A total of 27, 000 satellite images across 10 distinct land-cover classes were used for validation, ensuring diverse and comprehensive data coverage. The model achieved a peak accuracy of 97.5%, demonstrating robust performance and real-time detection capability. These findings highlight the strong potential of the YOLOv8 architecture for environmental monitoring, forest conservation, and sustainable resource management. Overall, this study represents an important step toward integrating artificial intelligence and satellite imagery for efficient and scalable forest observation.

Signature of the student(s)

Signature of the Guide

Signature of the project coordinator

